MATERIAL SAFETY DATA SHEET (MSDS)

Standard GHS Compliance Document for Laboratory Safety

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Chemical Name	Sulfuric Acid, 98%
Chemical Formula	H2SO4
CAS Number	7664-93-9
Document Purpose	Academic RAG Validation Dataset

SECTION 2: HAZARDS IDENTIFICATION

🔥 Severe skin burns and eye damage. Harmful if swallowed. Highly corrosive to metals. Strong dehydrating agent.

SECTION 4: FIRST AID MEASURES

- **Inhalation:** Move to fresh air. If breathing is difficult, give oxygen. Get immediate medical attention.
- **Skin Contact:** Immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing. Get medical attention instantly.
- **Eye Contact:** Immediately flush eyes with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Get medical aid immediately.
- **Ingestion:** Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth thoroughly with water and drink plenty of water afterwards.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- **Precautions:** Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Provide adequate ventilation.
- **Containment:** Stop leak if without risk. Absorb with dry earth, sand or other non-combustible material. Do NOT use water directly on the leak to prevent violent exothermic reactions.
- **Neutralization:** Neutralize carefully with lime (calcium oxide) or soda ash (sodium carbonate).

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Property	Value
Appearance	Oily, colorless to slightly yellow liquid
Odor	Odorless
pH	< 0.3 (1 N solution)
Boiling Point	337 °C (639 °F)
Melting Point	10.3 °C (51 °F)
Density	1.84 g/cm^3

SECTION 10: STABILITY AND REACTIVITY

Highly reactive. Exothermic dissolution with water (always add acid to water, never water to acid). Incompatible with organic materials, chlorates, carbides, fulminates, and picrates.

SECTION 16: OTHER INFORMATION

This document is synthetically generated for testing Retrieval-Augmented Generation (RAG) system pipelines within clinical biochemistry and safety management training workflows.